

Mechanical and thermal properties of pressureless sintered lunar regolith simulant: Analysis of sintering parameters and pore structure

Youxu Zeng, Haoyu Lu, Sheng Jiang, Lizhi Zhang, Haihan Tong, Ruifu Tang, Wenqiang Zuo^{*}

School of Materials Science and Engineering, Southeast University, Nanjing, 211189, China

ARTICLE INFO

Keywords:

Lunar regolith simulant
Pressureless sintering
Mechanical and thermal performances
Pore structure
Sintering kinetics

ABSTRACT

Sintering has emerged as one of the most promising techniques for in-situ resource utilization in lunar construction. Pressureless sintering, owing to its process simplicity and low equipment requirements, offers a feasible route for fabricating large and complex structural components. This study systematically investigates the effects of sintering temperature, holding time, and atmosphere on the mechanical properties, pore structure, thermal behavior, phase evolution, and microstructure of pressureless-sintered compacts using CUG-1B lunar regolith simulant. Results indicate that compacts sintered in air exhibit superior mechanical strength and reduced open porosity. Sintering temperature exerted a more significant influence on compact properties than holding time. Furthermore, sintering kinetics and pore-property relationships were explored, revealing that open pores (rather than isolated pores) predominantly degrade mechanical performance. In contrast, isolated pores significantly enhance thermal insulation. These findings provide critical insights and theoretical support for the application of pressureless sintering of lunar regolith in future large-scale extraterrestrial construction.

1. Introduction

In light of the rapid expansion of lunar exploration initiatives worldwide, such as the Artemis program and the Chang'e project [1,2], the construct of long-term inhabited lunar bases has become a pivotal objective in further deep space exploration [3,4]. However, constructing durable infrastructure on the Moon presents substantial challenges due to its extreme environmental conditions (ultra-high temperature variation, strong radiation, vacuum, low gravity, lack of water, etc.), as well as the high cost of Earth-Moon transportation [5]. Consequently, in-situ resource utilization (ISRU) has emerged as a critical strategy for lunar development, enabling the use of abundant local materials, particularly lunar regolith, as construction resources [6,7].

However, considering the scarcity of real lunar regolith (LR), systematic research into its production relies primarily on lunar regolith simulants (LRSs), such as HUST-1, KLS-1, JSC-2A, CUG-1 et al. [8–13]. These simulants are produced from terrestrial materials using tailored formulations and fabrication processes designed to replicate some of the characteristics of LR, including particle size distribution, mineral composition, particle shape, chemical composition, density, mechanical properties, and so on. Currently, three main approaches are employed

for the preparation and forming of LRSs: high-temperature solidification (sintering/melting) [14–17], chemical curing [18–20], and bonding [21–23]. Among these, sintering has attracted significant attention due to its procedural simplicity, ultra-high in-situ resources utilization efficiency (up to 100 %), and superior performance.

To date, extensive research has been conducted on the sintering behavior of LRSs. These studies primarily focus on employing various sintering processes (sintering temperature, holding time, heating rate, sintering atmosphere, etc.) [10,24–27] and energy methods (pressureless sintering, hot-press sintering, microwave sintering, spark plasma sintering, etc.) [11,16,27–29] to fabricate structures with improved mechanical strength or enhanced thermal insulation. The resulting sintered compacts are typically evaluated through a series of analyses, covering macro-level properties (compressive strength, Vickers hardness, etc.) [8,9,16,30], thermal properties (thermal conductivity, thermal expansion coefficient, etc.) [9,31–33], phase composition and microstructural evolution [8,25,26,26,33], as well as related numerical simulations and theoretical modeling [34–37].

However, systematic and multi-parameter comparative studies (involving temperature, soak time, and atmosphere) for pressureless sintered LRS (pressureless sintering, no external auxiliary pressure is

^{*} Corresponding author.

E-mail addresses: 230248400@seu.edu.cn (Y. Zeng), 230238734@seu.edu.cn (H. Lu), 230238738@seu.edu.cn (S. Jiang), 230228658@seu.edu.cn (L. Zhang), 213223257@seu.edu.cn (H. Tong), 213223651@seu.edu.cn (R. Tang), wengqiangzuo@seu.edu.cn (W. Zuo).

<https://doi.org/10.1016/j.ceramint.2025.12.293>

Received 18 September 2025; Received in revised form 3 December 2025; Accepted 17 December 2025

Available online 19 December 2025

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applied during the sintering process) remain scarce. Moreover, both theoretical and experimental studies on ceramic sintering indicate that the “connectivity/isolation of pores” determines their impact on mechanical and thermal properties (open pores significantly weaken mechanical strength, while isolated closed pores provide thermal insulation without substantially reducing strength). Although this theory is well-established in ceramic materials, literature directly quantifying the relationship between pore morphology and macroscopic mechanical/thermal properties remains scarce, especially for materials (such as the LRSs) with complex actual mineral compositions and coexisting glass/mineral phases.

In this study, sintered compacts were fabricated from the CUG-1B lunar regolith simulant (CUG-1B LRS) under varying processing conditions (sintering temperature, holding time, and atmosphere). Based on it, the macroscopic morphology, volume deformation, mass loss, volume change, porosity, and compressive strength were employed to analyze the macrostructure and mechanical performance evolution under sintering. Besides, the X-ray diffraction (XRD) and scanning electron microscope (SEM) were utilized to investigate the phase transformation and microstructure evolution under sintering. In addition, we further analyzed the relationship between mechanical performance and the sintering parameters, as well as the effect of pores on the mechanical and thermal performance of sintered compacts, by combining the above results and the data of other scholars.

2. Materials and methods

2.1. Raw materials

The LRS used in the experiment is CUG-1B (CUG-1B LRS), prepared by the Institute of Planetary Science at China University of Geosciences in Wuhan, China. It consists of plagioclase, monoclinic pyroxene, plagioclase pyroxene, olivine, ilmenite, quartz, magnetite, apatite, etc. Table 1 shows the main chemical composition and density of the CUG-1B LRS. It mainly consists of SiO_2 , Al_2O_3 , FeO , CaO , and MgO , accompanied by small amounts of TiO_2 , Na_2O , K_2O , P_2O_5 , etc. The apparent density is approximately 3.077 g/cm^3 (as determined by the specific gravity flask method). Fig. 1 illustrates the TG-DSC curve (a), particle size distribution (b), X-ray diffraction (XRD) curve (c), and scanning electron microscope (SEM) image (d) of the CUG-1B LRS powder. It can be seen that the particle size distribution of the CUG-1B LRS used in this study primarily ranges from 5 to $175 \mu\text{m}$, closely resembling the particle size distribution of LR power (Fig. 1a). Besides, the TG-DSC curves (Fig. 1b) reveal that the CUG-1B LRS exhibited minimal mass change below 550°C with no distinct endothermic/exothermic peaks, indicating good thermal stability. Significant mass loss occurred between 550 and 700°C , likely associated with the reduction of certain iron oxides within the mineral or the de-hydroxylation of hydroxyl-bearing minerals. Subsequently, it exhibited a sustained, gradual weight loss between 700 and 1000°C , associated with the relaxation of silicate structures within the minerals. Between 1000 and 1200°C , the mass decreased sharply again, corresponding to the softening of the glass phase and the formation of a molten liquid phase. After 1200°C , a distinct endothermic peak reappeared, indicating that the sample had essentially melted completely. This indicates that the suitable sintering range for CUG-1B LRS lies between 1000 and 1200°C . Meanwhile, the XRD spectrum (Fig. 1c) of CUG-1B LRS indicated that the crystal phases

mainly consisted of clinoclase, amphibole, plagioclase, and pyroxene, with a small amount of ilmenite and fayalite. It can also be seen that the particle shapes of the CUG-1B are mainly angular and subangular, containing a small amount of subcircular particles (Fig. 1d).

2.2. Sample preparation and sintering process

CUG-1B LRS powder was pressed by a dry powder compactor to obtain a cylindrical specimen with a size of about $\Phi 20 \times 20 \text{ mm}^2$ before sintering (Fig. 2 shows the appearance of the sample after pressing as well as the preparation process). Here, the weight of each sample is approximately 14.30 g , the maximum pressure applied was 100 MPa , and the maintenance time was 30 s . After that, the samples were placed in a tube furnace for sintering. The sintering atmosphere includes air and argon. The sintering temperatures are 1125°C , 1150°C , and 1175°C , and the holding times are set to 60 min , 180 min , and 300 min (identified by pre-experiment). Fig. 3 gives the sketch of the parameter settings for the various stages of CUG-1B LRS sintering. Besides, circular flakes with a thickness of about 2 mm and a diameter of about 20 mm were made using the same process for thermal property measurement.

2.3. Characterization methods

To reveal the macroscopic performance and structure of sintered compacts, the macrostructure, mass loss, structural deformation, volume change, porosity, mechanical property (compressive strength), X-ray diffraction, scanning electron microscope, and thermal conductivity of specimens sintered under different parameters are measured.

2.3.1. Structural deformation and volume change

The ratio $(\Delta D/H)$ between the bottom-top difference in diameter ($\Delta D = R_2 - R_1$) and height (H) of sintered compacts was applied to describe the degree of structural deformation. Besides, the volume change (ΔV) was measured using vernier calipers and then calculated by Eq. (1).

$$\Delta V = (V_1 - V_0) / V_0 \quad (\text{Eq. 1})$$

Where V_1 is the volume of the sintered compact and calculated by the formula $\pi H_1(R_1^2 + R_2^2 + R_1R_2)/3$; V_0 is the volume of the compacts before sintering and calculated by the formula $\pi H_0R_0^2$; H_1 is the height of sintered compacts, R_1 is the top surface radius of the sintered compacts, and R_2 is the bottom surface radius; R_0 is the radius of the compacts before sintering, and H_0 is the height.

2.3.2. Porosity

The open porosity (φ_{Open}) of the sintered compacts was measured using the water absorption method. After sintered, the mass (m_d) of sintered compacts was measured using an analytical balance. Then, the compacts were immersed in deionized water for 48 h , after which their mass (m_s) was measured. The formula for calculating the open porosity is shown in Eq. (2). Besides, the apparent density (ρ_{Apparent}) was measured by the specific gravity flask method to determine the total porosity (φ_{Total}). Eq. (3) presents the formula for the total porosity in sintered compacts.

$$\varphi_{\text{Open}} = \frac{(m_s - m_d) / \rho_{\text{water}}}{V} \quad (\text{Eq. 2})$$

Table 1

Chemical composition of CUG-1B LRS and the real LRs [38,39].

Oxides	SiO_2	Al_2O_3	TiO_2	$\text{FeO/Fe}_2\text{O}_3$	CaO	MgO	Na_2O	K_2O	P_2O_5
CUG-1B	45.64	13.40	0.17	9.22	7.80	9.16	3.00	1.90	0.49
Apollo-14	47.30	17.10	1.84	10.15	10.65	9.00	0.65	0.53	0.50
Apollo-17	45.20	20.10	1.40	8.50	12.50	10.0	0.44	0.17	
CE-5	42.20	10.80	5.00	22.5	11.00	6.48	0.26	0.19	0.23

Table 2

Explanation of labeling against crystals in Fig. 4.

Laber	1	2	3	4	5	6	7	8
Phase	Clinocllore	Amphibole	Plagioclase	Augite	Olivine	Ilmenite	Hematite	Magnetite

$$\varphi_{Total} = \left(V \frac{m_d}{\rho_{Apparent}} \right) / V \quad (\text{Eq. 3})$$

Where ρ_{Water} is the density of the deionized water (1 g/cm³ was taken in this study) and V is the volume of sintered compacts.

2.3.3. Compressive strength

Compressive strength was used to analyze the effect of sintering parameters on the mechanical properties of sintered lunar regolith simulant. In this study, the testing equipment employed was a computer-controlled electronic universal testing machine model UTM5105 (100 kN), with a loading rate of 100 N/s.

2.3.4. X-ray diffraction

XRD was used to analyze the crystalline phase composition and content of CUG-1B LRS before and after sintering, as well as the effect of sintering parameters on the mineral phase composition of the sintered compacts. In this study, the XRD patterns were recorded using a Rigaku SmartLab SE diffractometer with a Cu K α radiation source (40 kV and 40 mA) over the range of 5°–65° 2 θ with a step size of 0.02° 2 θ and a count time of 1°/min.

2.3.5. Scanning electron microscope

SEM was applied to investigate the effect of sintering parameters on the microstructure of the sintered compacts. The morphology of the compacts was acquired by means of Zeiss Sigma 300 SEM at an accelerating voltage of 20.0 kV and a working distance of approximately 10 mm, employing a secondary electron (SE) detector under high-vacuum mode. Given the poor electrical conductivity of the samples, they were sputter-coated with a thin conductive layer (~10 nm Pt) before observation to minimize charging effects.

2.3.6. Thermal conductivity

Thermal conductivity was measured by the flash thermal conductivity analyzer. Before testing, the sintered compacts were prepared as circular thin sheets approximately 12.7 mm in diameter and 2 mm thick, and subsequently tested using a Netzsch LFA 467 thermal conductivity analyzer to determine their thermal conductivity. Besides, considering the significant temperature variations on the lunar surface, the test temperatures were selected as follows: –100 °C, –40 °C, 20 °C, 80 °C, and 160 °C.

3. Results and discussion

3.1. Phase evolution

For the green sample (unsintered CUG-1B LRS), the primary crystalline phases consist of clinocllore, amphibole, plagioclase, and pyroxene, along with minor phases such as olivine, ilmenite, magnetite, etc. (Fig. 4). After sintered, the diffraction peaks corresponding to amphibole and clinocllore disappear, indicating their decomposition at high temperatures (Fig. 4). The crystalline phases in the sintered CUG-1B LRS are predominantly plagioclase (anorthite and albite) and augite, accompanied by smaller amounts of olivine and magnetite (Fig. 4). In comparison, only limited influence on the types of major crystalline phases, although they do affect their relative abundances (Fig. 4). Notably, specimens sintered in air contain small amounts of hematite (Fe₂O₃) and magnetite (Fe₃O₄), whereas only magnetite is present in those sintered under argon, with no detectable hematite

(Fig. 4).

The XRD spectra were quantitatively analyzed using the full-spectrum fitting method, and the quantitative results are presented in Fig. 5. After sintering, the amorphous content of the CUG-1B LRS decreases markedly, while the crystalline phases become dominated by plagioclase (albite, anorthite), augite, and olivine (forsterite, fayalite), along with minor amounts of ilmenite and magnetite (or hematite) (Fig. 5). This indicates that portion of the amorphous phase in the raw materials crystallizes during sintering. Besides, compared with argon-sintered CUG-1B LRS, more plagioclase and less olivine, augite, and amorphous phases are formed in air-sintered CUG-1B LRS.

As the sintering temperature increases and the holding time extends, the plagioclase content in the sintered CUG-1B LRS gradually decreases while the augite content increases (Fig. 5). The reduction in plagioclase content may be attributed to the formation of more melt surrounding its grains within this temperature range, as well as solid solution decomposition and localized melting [41,42]. The increase in augite content is likely associated with phase transformations triggered by the melting of plagioclase, resulting in the formation of enstatite and other minerals [41,43].

3.2. Macroscopic morphology

In this study, a digital camera and a high-definition electron microscope were used to examine the macroscopic structure of the sintered compacts. Fig. 6 shows the macroscopic morphology and low-magnification optical micrographs of sintered CUG-1B LRS compacts under different thermal processes. Compacts sintered in air exhibit a reddish-brown color (Fig. 6a–e), whereas those sintered in argon appear gray-brown (Fig. 6f–j), consistent with the observations of Song et al. [44]. Meanwhile, the color of the compacts darkens with increasing sintering temperature: the compacts sintered in air range from light red to reddish-brown, while those sintered in argon range from gray to gray-brown. A similar trend is observed when holding time is increased, with the compacts becoming progressively darker under both atmospheres.

The reddish-brown color of air-sintered compacts can be attributed to the oxidation of ilmenite (Figs. 4 and 5), during which Fe²⁺ is oxidized to Fe³⁺ to form Fe₂O₃ (hematite) (4FeO + O₂ → 2Fe₂O₃ (red-dish-brown)) [45]. In contrast, under argon sintering conditions, the absence of oxygen promotes reduction. Consequently, the Fe³⁺ in ilmenite undergoes a disproportionation reaction (6Fe₂O₃ → 4Fe₃O₄ (gray-brown)), resulting in the formation of magnetite (Fe₃O₄) (Figs. 4 and 5) with an antiferroelectric spinel structure [45]. Meanwhile, the rock-salt-type structure of galena (FeO) strongly absorbs the full-wave band of visible light. The combined effects of these two phases impart a gray-brown color to the surface of argon-sintered compacts [46]. Besides, according to the Arrhenius theory [47], changing the sintering temperature significantly alters the reaction activation energy, leading to a shift of the absorption peaks, resulting in a darker color in the compacts sintered under higher temperatures [48]. Additionally, enhanced formation of glassy at elevated sintering temperatures or prolonged holding times reduces the refractive index contrast with oxides. This suppresses light scattering and surface reflectivity according to Fresnel theory [49,50], further darkening the appearance. Additionally, the higher temperature and the longer time also promote the grain coarsening (Fig. 7) and densification (Fig. 9). Larger-sized grains absorb light more efficiently, while the densification of the sintering helps to reduce surface light scattering [51]. Together, these factors contribute to the darker coloration observed in the sintered compacts.

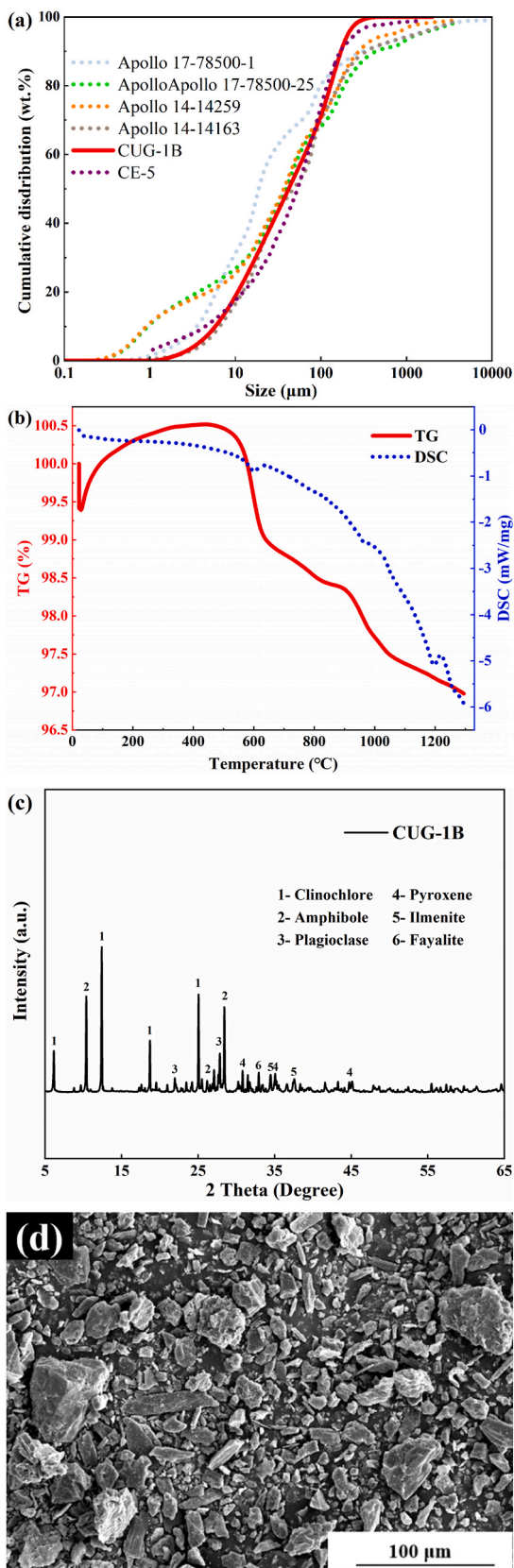


Fig. 1. Characterization of raw material. (a) Particle size distribution (CUG-1B LRS and the LR [39,40]); (b) X-ray diffraction curve; (c) TG-DSC curve; (d) Scanning electron microstructure.

Furthermore, magnified images reveal that the surfaces of air-sintered compacts contain numerous microcracks (Fig. 6a3~e3), whereas argon-sintered compacts show no visible microcracking (Fig. 6f3~j3). This difference is likely due to stresses induced by oxidation during sintering in air [52,53]. Additionally, the air-sintered compacts at 1175 °C for 60 min (Fig. 6a1) and 1150 °C for 300 min (Fig. 6e1) present a truncated conical shape. This deformation may result from the excess liquid phase formed at the higher temperature and longer holding time, which flows under the influence of gravity, resulting in shape changes [54,55].

3.3. Microstructure evolution

Fig. 7 presents the micromorphology of CUG-1B LRS compacts sintered under different thermal conditions. As the sintering temperature increases, the microstructure becomes denser, the pores decrease, and the particles are more tightly bonded. A similar trend is observed with increasing holding time, which also promotes densification. Additionally, the compacts exhibit larger grain size at higher temperature (such as Fig. 7f~j) and longer holding time (such as Fig. 7a and b). This is consistent with the traditional sintering kinetics and microstructural evolution: prolonging the holding time facilitates continued diffusion and liquid-phase infiltration, which increases the sintering neck width and significantly reduces the submicron pores due to dissolution-precipitation, while the crystalline phases precipitate in the glass matrix [56,57].

It is also found that the microstructure of the sintered compacts in the air exhibited greater density, lower porosity, and smaller pore sizes compared to those sintered in argon (such as Fig. 7c~h), as the sintering temperature and holding time were consistent. Moreover, the grain size of the compacts sintered in air is significantly larger than that of the compacts sintered in argon (such as Fig. 7b~g). These differences may be attributed to variations in apparent activation energy, viscosity, liquid phase content, and other properties during sintering under the 2 atm, which in turn affect the densification behavior during sintering.

Meanwhile, a distinct solidified liquid phase was observed in compacts sintered in air at higher temperatures (Fig. 7c) and longer holding time (Fig. 7e), whereas such features were absent in those sintered in argon. As the temperature rises, the oxidation reaction accelerates, the contact area between the particles increases quickly, and the silicate partially melts, forming a continuous glass phase that fills the pores and surrounds the particles [58]. This also led to the pores becoming smaller or even isolating.

3.4. Geometry, mass, and volume change

The ratio between bottom-top difference (ΔD) and height (H) of sintered compacts (Fig. 8a and b) further confirms that their shape evolves from cylindrical to the shape of a frustum of a conical structure with increasing temperature and holding time. The change in the degree of structural deformation of the sintered compact with increasing sintering temperature may be due to the evolution of the phase equilibrium of the multiphase system. When the temperature surpasses the eutectic point of silicate minerals, the glassy phase increases exponentially. Once the liquid phase fraction exceeds 30 %, gravitational acceleration ($g = 9.8 \text{ m/s}^2$) induces Stokes' flow, producing a compositional gradient along the gravity direction [54,55]. During subsequent cooling, the difference in thermal expansion coefficients between the glass phase and residual crystalline phases (for example, $\alpha_{\text{glass}} \approx 8.5 \times 10^{-6}/^\circ\text{C}$ and $\alpha_{\text{ilmenite}} \approx 10 \times 10^{-6}/^\circ\text{C}$) [31,32] generates thermal stresses in localized regions that exceed the material's yield strength, ultimately inducing plastic deformation. Besides, elevated temperatures promote grain growth [48], and oversized grains render the microstructure inhomogeneous and susceptible to deformation [59].

Except for the sintering atmosphere, the sintering temperature and holding time have a negligible influence on the mass loss ($\Delta m/m$, mass

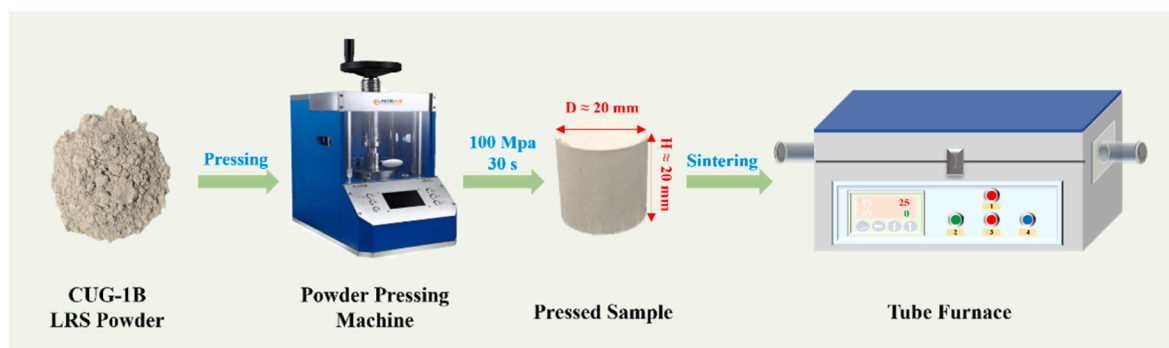


Fig. 2. Sketch of the sample preparation and sintering process.

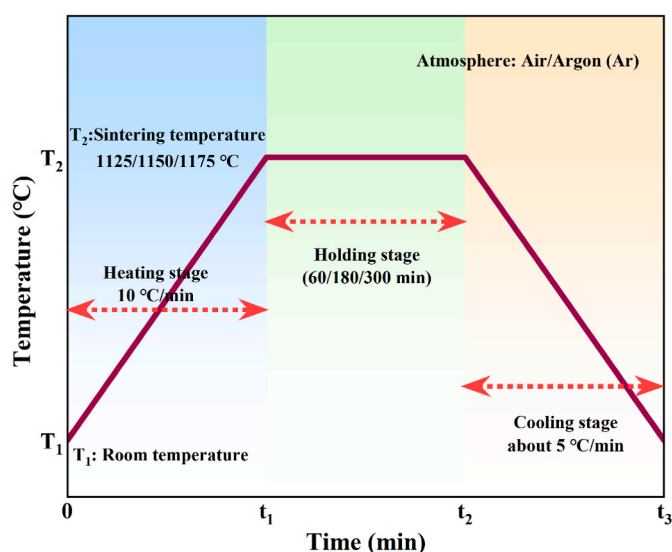


Fig. 3. Sketch of the parameter settings for the various stages of CUG-1B LRS sintering.

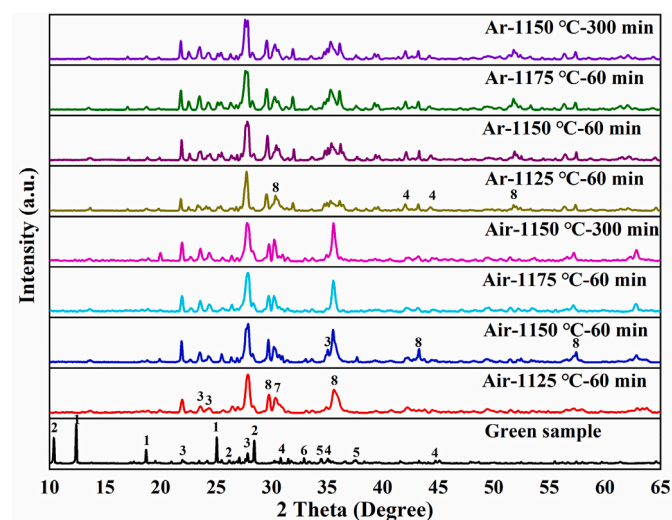


Fig. 4. X-ray diffraction spectra of CUG-1B LRS before and after different thermal sintering processes. (The crystal diffraction peaks corresponding to each label are provided in Table 2.)

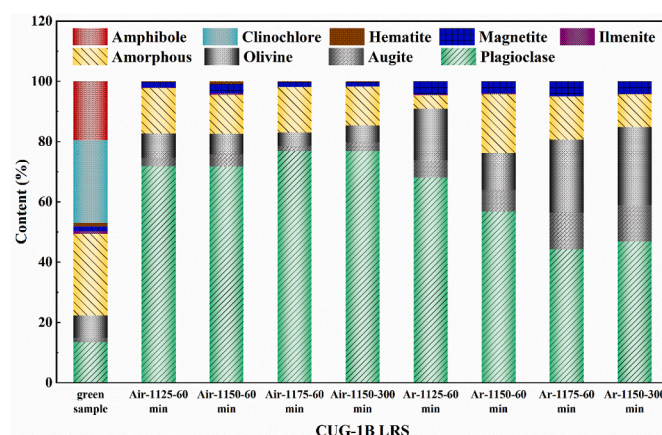


Fig. 5. Content of major crystals and amorphous.

difference variation between samples before and after sintering) of sintered compacts (Fig. 8a and b). However, the compacts sintered in the air appear to have a lower mass loss (lower than 1 %) compared to the compacts sintered in argon (Fig. 8a and b). This difference arises from the behavior of minerals in the raw material (such as ilmenite). In air, sufficient oxygen promotes Fe^{2+} oxidation in ilmenite (FeTiO_3) above 600 °C, causing a 4.8 % mass gain per mole that partially offsets volatilization loss [45]. In argon, oxygen deficiency drives ilmenite reduction at high temperatures to FeO or Fe_3O_4 , releasing oxygen and producing a 0.8 % mass decrease [45]. Combined with volatilization, this leads to a greater apparent mass loss for argon-sintered compacts.

The volume of sintered compacts (Fig. 8c and d) exhibits an initial expansion followed by contraction with increasing temperature, while the effect of holding time is minimal. Meanwhile, it is evident (Fig. 8c and d) that the magnitude of the volume change is greater for compacts sintered in air than for those sintered in argon as the temperature rises. Compact expansion may occur during the early sintering stage as volatile decomposition releases gases that, if trapped, induce swelling. In addition, competition between particle rearrangement and insufficient energy for stable contact formation can cause expansion [48]. Additionally, mineral transformation during sintering also leads to expansion. For example, the SiO_2 undergoes α - β transition at 573 °C (with 0.8 % expansion) and further transforms into squamous quartz or cubic quartz above 870 °C (with 15 % expansion) [60,61]. At higher temperatures, partial melting generates liquid phases that wet particles, fill pores, and facilitate rearrangement, leading to densification and shrinkage [55]. Simultaneously, grain boundary and volume diffusion accelerate material transport, sintering neck growth, and grain consolidation, thereby reducing porosity and overall volume [55].

The difference in volume change of sintered compacts in air and argon at elevated temperatures can be attributed to oxidation in the

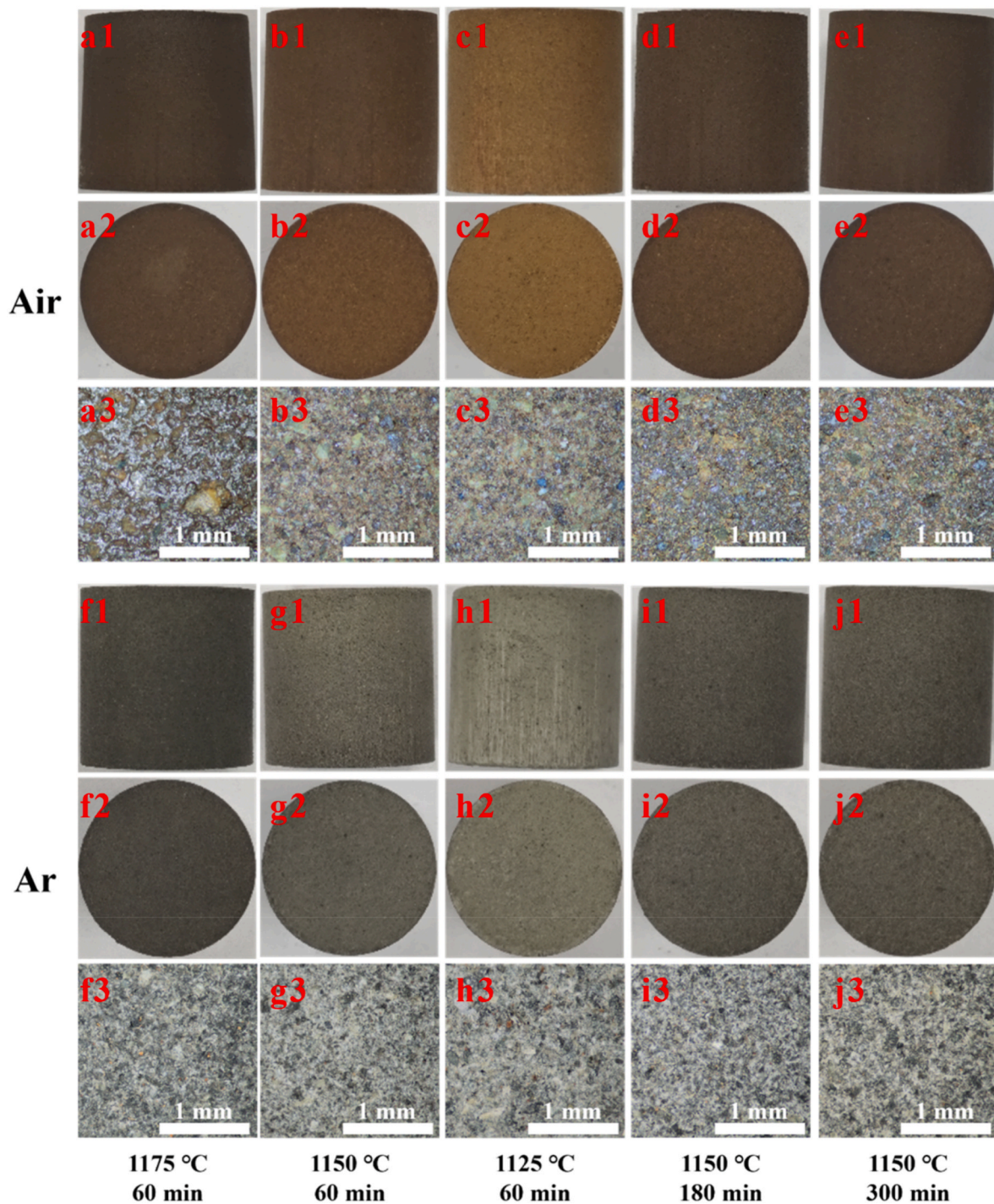


Fig. 6. Macroscopic morphology and low-magnification optical micrographs of sintered CUG-1B LRS compacts under different thermal sintering processes (the diameter and height of the cylinder are approximately 20 mm).

presence of oxygen [62]. Oxidation promotes the formation of additional liquid phases during sintering, accelerating the densification and leading to greater shrinkage. Conversely, the leakage of oxygen in argon restricts liquid phase formation, leading to a lower degree of shrinkage. This effect becomes more pronounced at higher temperatures, where oxidation is more intense, generating more liquid phase and thus greater shrinkage. Moreover, the weak influence of holding time on volume change may be owing to the limited content of minerals with low melting temperature [55].

3.5. Compressive strength and porosity

Compacts sintered at higher temperatures exhibit superior compressive strength and lower porosity (open porosity and total porosity) (Fig. 9a). Similarly, sintered compacts exhibit higher compressive strength and lower porosity with increased holding time (Fig. 9b). By contrast, the effect of holding time on the improvement in strength and decrease in porosity was less pronounced compared to the sintering temperature (Fig. 9a and b). Besides, the compressive strength of the compacts sintered in air is much higher than those sintered in argon, as the sintering parameters are consistent (Fig. 9a and b).

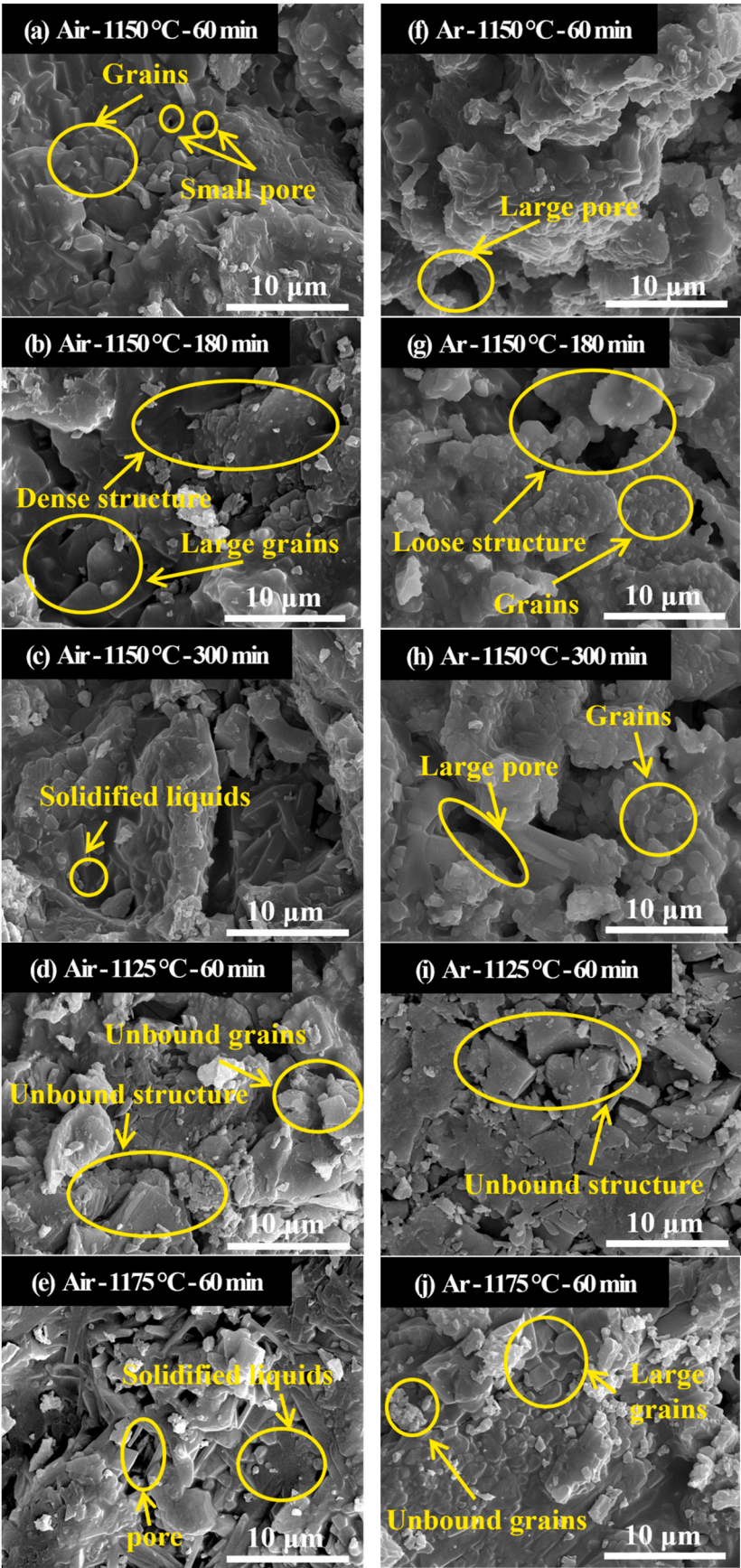


Fig. 7. SEM of sintered CUG-1B LRS compacts after different thermal sintering processes.

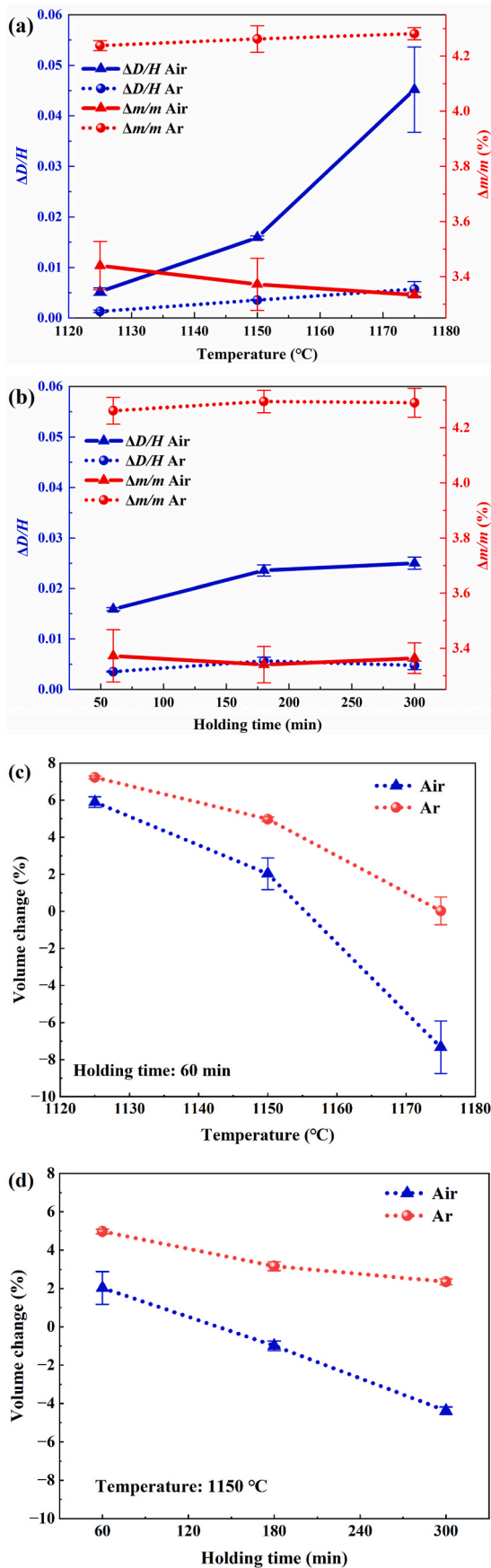


Fig. 8. Structural deformation (a, b), mass loss (a, b), and volume change (c, d) of sintered CUG-1B LRS compact under different thermal sintering processes.

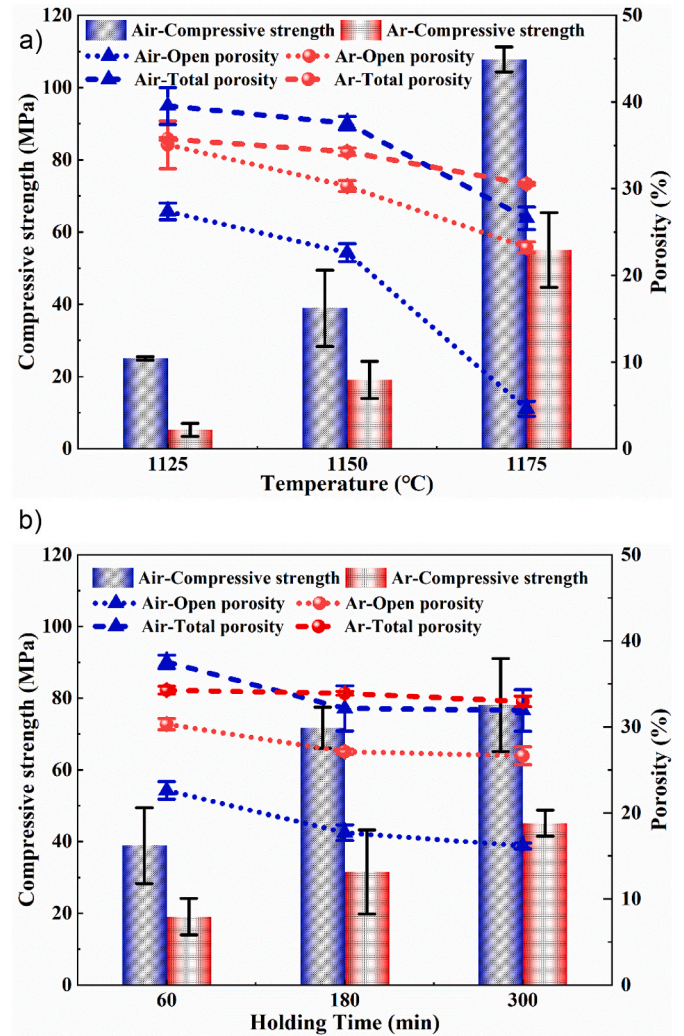


Fig. 9. Compressive strength and porosity of sintered CUG-1B LRS compacts under different thermal sintering processes. (a) Hold at different sintering temperatures for 60 min, and (b) Hold at 1150 $^{\circ}\text{C}$ for different times.

The superior compressive strength and lower porosity with higher temperatures can be attributed to the synergistic effect of multiscale physicochemical processes at high temperatures. First, increasing temperature markedly accelerates atomic and ionic diffusion on particle surfaces (consistent with Arrhenius theory), promoting material migration at contact regions, rapid sintering neck growth, pore closure, and stronger interparticle bonding [55]. Concurrently, partial melting of silicate minerals (e.g., plagioclase, pyroxene) produces glassy or liquid phases that wet particles, infiltrate pores via capillary action, and solidify into a robust glassy bonding network upon cooling. High temperatures also trigger mineral phase transitions, including recrystallization of amorphous phases and transformation into more stable crystalline forms, thereby enhancing strength and reducing porosity through associated pore volume changes [58].

The primary mechanism that substantially enhances the compressive strength and lowers the porosity of the sintered compacts by prolonging the holding time originates from the optimization effect of physicochemical processes at elevated temperatures [56]. With increasing holding time, cumulative atomic diffusion drives continuous sintering neck growth, enlarging particle bonding areas and strengthening interparticle forces. Simultaneously, silicate liquid phase infiltrates the pore network via capillary action, eliminating small pores through dissolution and precipitation [58]. Besides, the progressive formation of crystalline phases such as augite and olivine further enhances structural

integrity, while residual stress relaxation and pore rounding mitigate stress concentration [58].

The finding that compacts sintered in air exhibit higher compressive strength and lower open porosity than those sintered in argon can be attributed to the multiple optimizations of the microstructure by the oxidation reaction [63]. Oxidation in air promotes sintering by generating hard oxides that react with silicates to form high-strength crystalline phases, reinforcing the skeleton and filling pores. The exothermic oxidation induces local temperature rises that promote atomic diffusion and neck growth. Furthermore, Fe^{3+} formation modifies the melt structure, accelerates silicate polymerization, and yields a glassy phase with a higher elastic modulus.

3.6. Thermal property

The thermal conductivity under different temperatures for the sintered CUG-1B LRS compacts was measured (Fig. 10). The results show that the thermal conductivity of the compacts increases with rising test temperatures ($-100\sim 80\text{ }^{\circ}\text{C}$) and subsequently stabilizes as the temperature exceeded $80\text{ }^{\circ}\text{C}$. This is consistent with the trend of phonon scattering weakening as temperature decreases in porous ceramics. Interestingly, compacts sintered at higher temperatures and longer holding times exhibited a lower conductivity compared to other compacts. Typically, raising the sintering temperature or lengthening the holding time leads to the densification of the sintered compacts, a decrease in porosity, and an increase in the number of thermal conduction pathways, thus raising the thermal conductivity. This seems to conflict with our results, suggesting that other important factors also affect the heat performance of sintered compacts.

Furthermore, under identical sintering conditions ($1150\text{ }^{\circ}\text{C}$, 300 min), the air-sintered compacts possess lower thermal conductivity than those sintered in argon. This difference may arise from the influence of the sintering atmosphere on both densification and the composition and proportion of mineral phases. As shown in Fig. 9b, sintered compacts Air- $1150\text{ }^{\circ}\text{C}$ -300 min and Ar- $1150\text{ }^{\circ}\text{C}$ -300 min have similar total porosity, yet their open pore content differs significantly. This suggests that the ratio of open to isolated pores affects the thermal conductivity of sintered compacts. Additionally, phase analysis (Figs. 4 and 5) also reveals differences in mineral phase composition and content between compacts sintered in the 2 atm.

Meanwhile, it can be seen that the thermal conductivity of the sintered compact Ar- $1125\text{ }^{\circ}\text{C}$ -60 min is much higher than that of other sintered compacts. However, SEM morphology (Fig. 7i) reveals that its

structure is significantly less compact compared to other compacts. This discrepancy may result from differences in the proportions of the various mineral phases formed at different temperatures, as well as variations in pore structure and size. Further analysis was conducted on the factors affecting the thermal property of the sintered compact in Section 4.4.

4. Correlation analysis of material properties governed by sintering

4.1. Sintering parameters and kinetics

Apparent activation energy (Q) results (Fig. 11) obtained from experiments based on Arrhenius theory (Eq. (4)) suggest that air-sintered CUG-1B LRS compacts have a much lower Q ($Q_{\text{Air}} \approx 312\text{ kJ/mol}$) than argon-sintered compacts ($Q_{\text{Ar}} \approx 627\text{ kJ/mol}$). This explains why better performance (Fig. 9) and higher structural deformation (Fig. 8a and b) are exhibited by air-sintered compacts under the same sintering parameters. Besides, it also explains the variation in grain size, pore content, and pore size of sintered compacts in air and argon (Fig. 7).

The linear shrinkage of the sintered compacts was experimentally measured as a function of holding time, with the results shown in Fig. 12. The data indicate that the linear shrinkage rate of the sintered compacts is initially large and decreases rapidly, continuously to decline gradually with further extension of holding time. Notably, the linear shrinkage-time curves of the two systems are essentially consistent at $1150\text{ }^{\circ}\text{C}$, revealing that sintering in both air and argon triggers a liquid-phase-dominated densification mechanism [64]. Which can be derived by combining the characteristics of the various stages of liquid phase sintering (Eq. (5)): (i) Particle rearrangement process is completed very quickly during sintering (holding time $<30\text{ min}$). Pressure compaction before sintering accelerated the process. Meanwhile, particle rearrangement occurred during the heating process. (ii) Subsequently, the compacts underwent dissolution-precipitation (holding time $<120\text{ min}$) ($\alpha \leq 1/2$). (iii) And then, as the holding time prolonged, the compacts entered the solid-phase sintering stage ($\alpha \leq 1/3$). Additionally, compacts sintered in argon underwent particle rearrangement and dissolution-precipitation faster than compacts sintered in air (Fig. 11), resulting in a higher open pore volume and a lower compressive strength (Fig. 9).

Meanwhile, given that the rate amplitude (K in Eq. (6)) is associated with solubility, diffusion coefficient, interfacial reaction coefficient, etc., the atmosphere (Air/Ar) exerts a predominant influence on the amplitude coefficient (K) and apparent activation energy (Q) by modifying the liquid phase fraction (ϕ_l), viscosity (η), contact angle (θ), etc. This

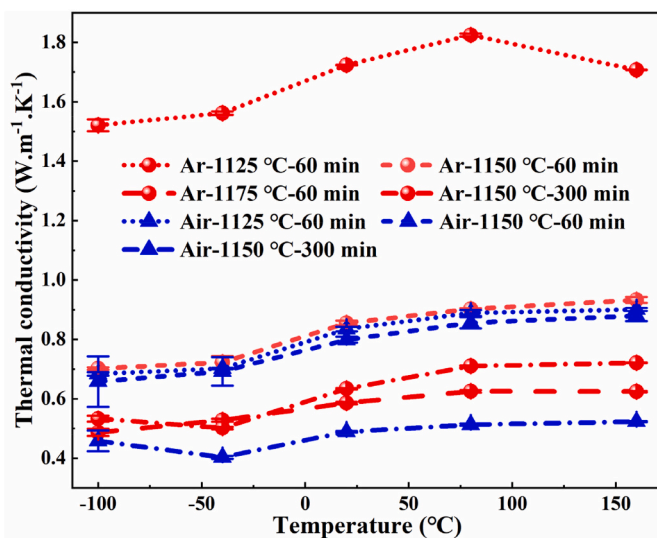


Fig. 10. Distribution of thermal conductivity under different temperatures for the sintered CUG-1B LRS compacts.

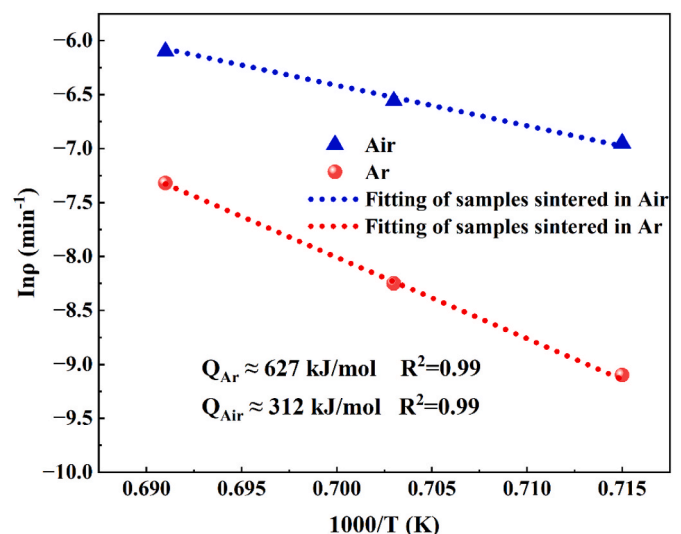


Fig. 11. Arrhenius plots for CUG-1B LRS compacts sintered in air and argon.

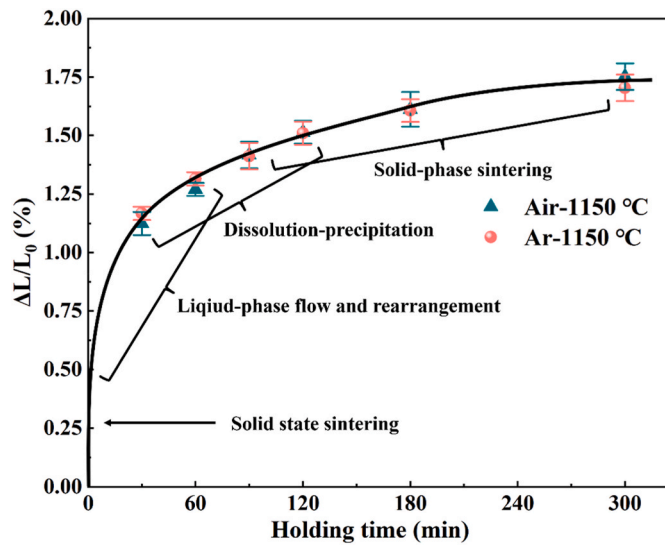


Fig. 12. Linear shrinkage as a function of time and the sketch of the various stages of liquid phase sintering [48].

results in differences in the apparent activation energy of the 2 atm (Fig. 12) and the overlap of the linear shrinkage-time curves (Fig. 12).

$$D = D_0 \exp\left(-\frac{Q}{RT}\right) \quad (\text{Eq. 4})$$

Where D is the rate constant or a quantity related to the sintering rate (in this study, the densification rate was selected); D_0 is the pre-exponential factor; Q is the apparent activation energy; R is the gas constant; T is the Kelvin temperature.

$$\frac{d(\Delta L/L)}{dt} = Kt^\alpha \quad (\text{Eq. 5})$$

$$K = K_0 \exp\left(-\frac{Q}{RT}\right) F(\phi_l, \theta, \eta, \gamma, \rho, r) \quad (\text{Eq. 6})$$

Where $\Delta L/L$ is the linear shrinkage; K is the rate amplitude; t is the time; α is the time exponent; K_0 is the pre-exponential factor; $F(\cdot)$ is the combination of geometric and physical properties; ϕ_l is the liquid phase fraction; θ is the contact angle; η is the viscosity; γ is the interfacial energy; ρ is the relative density; r is the particle/pore characteristic scale.

4.2. Sintering parameters and mechanical property

Fig. 13 gives the distribution of the compressive strength of sintered CUG-1B LRS compacts concerning the sintering temperature and holding time. It is evident that the sintering temperature is the dominant factor affecting the compressive strength of sintered compacts.

The effect of holding time on the compressive strength of the sintered compacts is significantly weaker than the effect of sintering temperature. Especially, the enhancement effect weakened considerably with the continuous extension of holding time. However, despite the higher sintering temperatures yielding compacts with better performance, they also lead to more significant dimensional deformation.

For future lunar construction, sintered compacts are more likely to be assembled in a modular manner to construct various structures, making excessive dimensional deformation undesirable. Additionally, given that lunar gravity is only one-sixth of Earth's gravity, sintered compacts with performance above 20 MPa can already meet the requirements for most construction applications. Accordingly, the results can be categorized into different areas (as present in Fig. 13) by considering the compressive strength, structural deformation, and sintering energy consumption

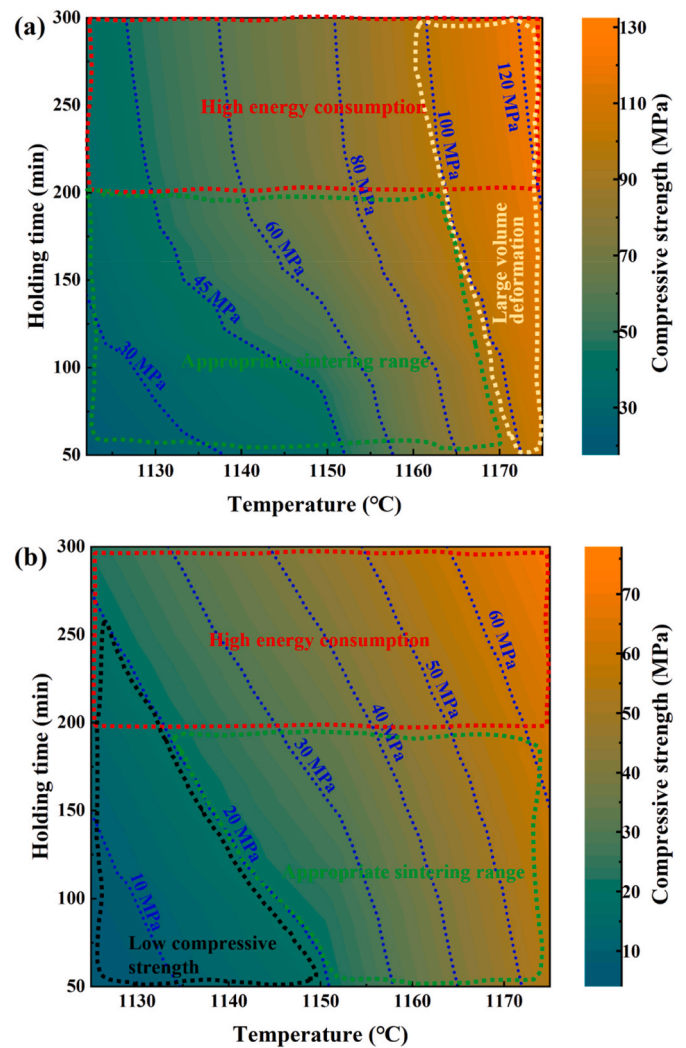


Fig. 13. Distribution of the compressive strength of sintered CUG-1B LRS compacts ((a) air atmosphere, and (b) argon atmosphere) concerning the sintering temperature and holding time.

and efficiency. Although the specific sintering parameters do not apply to other LRSs and LR due to differences in mineral composition and other factors, the changes in distribution that occur with holding time and sintering temperature are universal. For other ultra-high-performance sintered structures, alternative solutions can be prepared through methods such as optimizing the particle size distribution of sintering powders or employing pressure-assisted sintering.

It is also clear that, under identical sintering conditions, the compressive strength of compacts sintered in an argon atmosphere (Fig. 13a) is significantly lower than that of compacts sintered in an air atmosphere (Fig. 13b). To further investigate the effect of the sintering atmosphere on the sintered compacts, additional compacts were prepared in argon atmosphere at 1200 °C for 60 min, which exhibited noticeable melting. This indicates that the formation of oxygen helps promote the formation of the liquid phase. Consequently, compacts sintered in an air atmosphere with the same parameters exhibited higher performance. This also suggests that, in future lunar construction, an appropriate amount of oxygen could be introduced to produce ultra-high-performance sintered structural components.

4.3. Pore structure and mechanical property

Fig. 14 gives the compressive strength distribution along the total

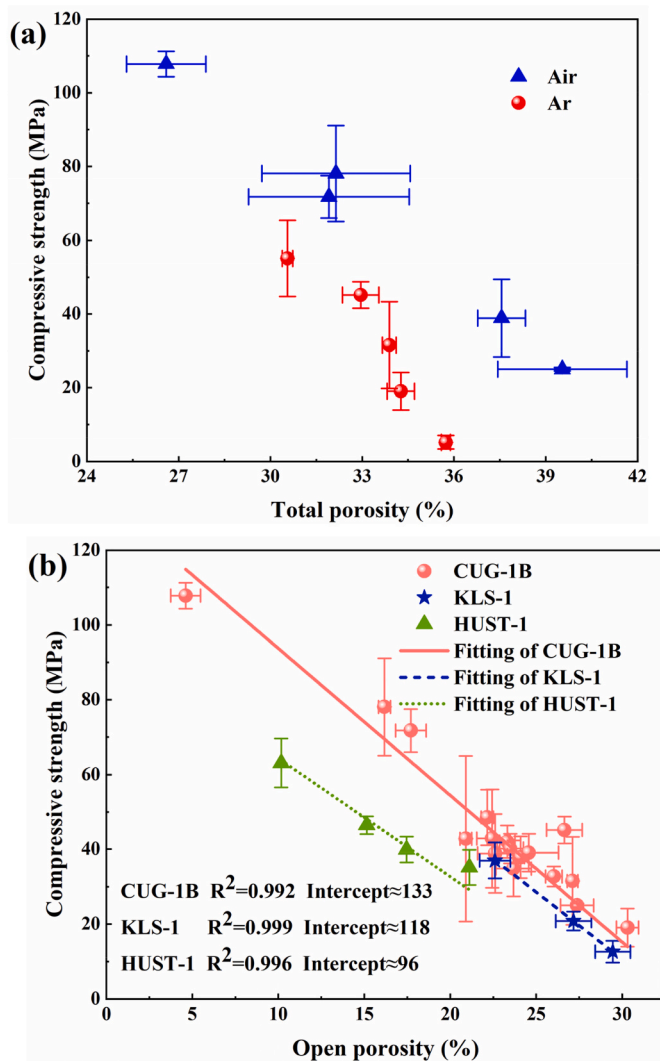


Fig. 14. Relationship between compressive strength and porosity of sintered compacts. (a) The compressive strength of sintered CUG-1B LRS compacts as a function of total porosity distribution. (b) The compressive strength of sintered LRSs compacts as a function of open porosity distribution and their fitting curves (the data for sintered KLS-1 and HUST-1 compacts are from references [27,33]).

porosity (Fig. 14a) and open porosity of sintered lunar regolith simulants (including the CUG-1B, KLS-1 (microwave sintering), HUST-1 LRS (spark plasma sintering)) [27,33] as well as their fitting curves (Fig. 14b). It demonstrates a negative correlation between the compressive strength and porosity.

Generally, the higher the porosity of the block, the lower its compressive strength. For ceramics, the most typical relationship between pore structure and mechanical properties is the Ryshkevitch–Duckworth model (Eq. (7)) [65]. However, our research results (Fig. 14a) indicate that, for the same material, the model cannot perfectly fit the sintered compacts when the sintering atmosphere is ignored. This suggests that other important factors affect the pore structure–mechanical relationship of ceramics.

Interestingly, the compressive strength and open porosity of the sintered compacts present a negative correlation (Fig. 14b). The fitting curve exhibits a consistent negative linear correlation (Eq. (8)), even across the different sintering conditions. To confirm this relationship further, we varied the pressing pressure, pressing time, and heating rate, and found that these factors did not affect its validity. Furthermore, data (Fig. 14b) from other studies, including microwave-sintered KLS-1 LRS

compacts and spark plasma sintered HUST-1 compacts (pre- and unpressed, and pre- and unpressed) [27,33], demonstrate the same relationship, with differences only in the slope (k value) in the relationship (Eq. (8)).

Considering the slope of fitting curves is adjacent to the intrinsic strength [66,67] of the natural mineral, the relationship between compressive strength and open porosity can be expressed by Eq. (9). This also indicates that the factors such as sintering technique (eg, traditional sintering, hot press sintering, microwave sintering), sintering atmosphere (air, argon, vacuum, etc.), as well as thermal sintering parameters have only a minor effect on the relationship. Furthermore, it also suggests that the relationship does not apply to isolated pores, as they are fully enclosed within the matrix and therefore introduce only local reductions in load-bearing cross-section without providing a continuous pathway for crack growth.

$$\sigma = \sigma_0 \exp(-bP) \quad (\text{Eq. 7})$$

Where σ is the compressive strength of the material; σ_0 is the theoretical strength of the material, or its compressive strength in the absence of pores; b is the material constant related to pore morphology, pore size, and distribution (usually 2 to 7); P is the total porosity of the material.

$$\sigma = k(1 - \varphi_{\text{Open}}) \quad (\text{Eq. 8})$$

$$\sigma = \sigma_0(1 - \varphi_{\text{Open}}) \quad (\text{Eq. 9})$$

Where σ is the compressive strength of sintered LRS; k is the slope; σ_0 is the intrinsic strength of LRS; φ_{Open} is the open porosity of sintered LRSs.

4.4. Pore structure and thermal property

Increasing the sintering temperature (Ar-1125 °C-60 min and Ar-1175 °C-60 min) or extending the holding time (Air-1150 °C-60 min and Air-1150 °C-300 min) leads to a decrease in total porosity of the sintered compacts, an increase in isolated porosity, and a decrease in thermal conductivity. In general, lower porosity in a material corresponds to higher thermal conductivity and poorer insulation performance. Nevertheless, our results reveal that the increase in isolated porosity not only curbs the rise in thermal conductivity caused by reduced porosity but also further undermines the thermal conductivity. This indicates that isolated porosity is the key factor governing the thermal conductivity as it effectively traps gas within fully enclosed cavities and interrupts continuous heat-transfer pathways through the solid matrix.

It also explains why the thermal conductivity of sintered compact Ar-1125 °C-60 min is much higher than that of other sintered compacts. Certainly, the effect of open pores on thermal conductivity cannot be ignored. When the isolated pore content is similar, the higher the open porosity, the better the thermal insulation performance (Ar-1150 °C-300 min and Ar-1175 °C-60 min).

Further analysis of Fig. 15 also reveals that the thermal conductivity of the air-sintered compacts is higher than that of the argon atmosphere-sintered compacts when the open porosity is similar, despite the higher isolated porosity of the air-sintered compacts (Air-1150 °C-60 min and Ar-1175 °C-60 min). This variation may be related to the different material transformation processes during the sintering process in the 2 atm. It also indicates that the minerals formed in air are more thermally conductive than the phase formed in argon.

5. Conclusions

In this study, the CUG-1B LRS was used to analyze the effect of sintering parameters on the macrostructure, mechanical performance, phase transition, and microstructure of pressureless sintered LRS compacts. Based on it, the relationship between mechanical performance and the sintering parameters, as well as the effect of pores on the mechanical performance of sintered compacts, was also discussed. The

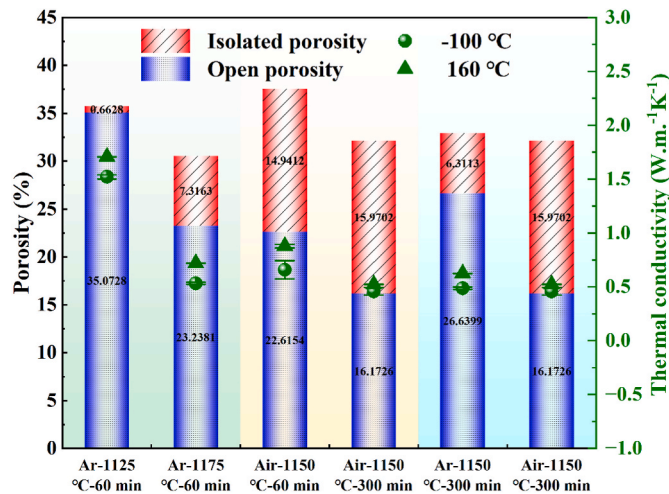


Fig. 15. Influence of open pores and isolated pores on the thermal conductivity of sintered CUG-1B LRS compacts.

following main conclusions can be drawn.

- (1) The oxygen-free environment exerts a detrimental effect on the mechanical properties, thermal properties, and pore structure of lunar regolith simulant produced by sintering. Compacts sintered in air exhibited notably higher mechanical strength and lower open porosity compared with those processed in inert atmospheres. Compacts sintered in air exhibited notably higher mechanical strength and lower open porosity compared with those processed in inert atmospheres.
- (2) Innovatively incorporates pore type into the performance evaluation system for sintered lunar regolith simulant, quantitatively expressing the relationship between their open porosity and mechanical properties. This provides new design guidelines for simultaneously optimizing load-bearing capacity and thermal insulation in future lunar structures.

Pressureless sintering plays an irreplaceable role in future lunar construction, which is a key step in the exploration of space. With lunar gravity being only 1/6 of Earth's, the strength of pressureless sintered compacts already meets the requirements of most applications, while also offering excellent thermal insulation properties. Moreover, our research findings indicate that moderately regulating sintering temperature and holding time to preserve a specific isolated porosity while minimizing open porosity is pivotal for achieving superior thermal insulation and mechanical characteristics of sintered compacts. Consequently, excessive densification should not be the primary objective. Rather, a harmonious balance must be sought between mechanical properties and thermal properties.

CRedit authorship contribution statement

Youxu Zeng: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Haoyu Lu:** Writing – review & editing, Formal analysis, Data curation. **Sheng Jiang:** Writing – review & editing, Data curation. **Lizhi Zhang:** Writing – review & editing, Data curation. **Haihan Tong:** Data curation. **Ruifu Tang:** Data curation. **Wenqiang Zuo:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Funding acquisition, Data curation, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used [ChatGPT 5.0/Deepseek] in order to improve the language and readability. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors gratefully acknowledge the financial support from the National Key Research and Development Program of China (No. 2023YFB3711300, No. 2021YFF0500300), Southeast University Interdisciplinary Research Program for Young Scholars (No. BK20241340) and the Fundamental Research Funds for the Central Universities (No. 2242025F10009).

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